

Candidate Seat No: _____

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Bachelor of Computer Applications(BCA)

Semester-I University Examination March-2018

CA 109 Introduction to Information Technology

Date: 26.03.18, Monday Time: 10.00 a.m. to 01.00 p.m.

Maximum Marks: 70

Instructions:

1. Attempt both sections in separate answer books.
2. The figure to the right indicates marks.
3. Make a suitable assumption when necessary.
4. Simple Calculator is permitted.

SECTION-1

Q.1 [A] Match column A with the correct answer on column B. [05]

Column A

1. NIU
2. Archival Store
3. Polycarbonate storage cell
4. DRAM
5. ROM

Column B

- a. Laser Disk or Compact Disk
- b. Capacitor
- c. Long life
- d. Non volatile
- e. Connect to communication network

[B] Give an appropriate answer to the following multi choice questions. [06]

1. The area required to fabricate capacitor storage is _____.
(a) 0.02cm^2 (b) $0.01\mu\text{m}^2$ (c) 0.01cm^2 (d) $0.02\mu\text{m}^2$
2. _____ Processor is required to process audio and video data.
(a) Digital Signal Processor (b) NIC (c) NIU (d) All of these
3. Map is an example of _____ type of data.
(a) Textual (b) Numeric (c) Audio (d) Image
4. In _____ ROM data cannot be erased by any means.
(a) EPROM (b) PROM (c) EEPROM (d) None of these
5. _____ system of computer acquires data from the external world.
(a) Output (b) Processing (c) Input (d) Memory
6. ASCII uses _____ bits to code each character.
(a) 9 (b) 8 (c) 7 (d) 2

Q.2 [A] Explain different types of Data with at least two examples of each type. [05]
Differentiate Data and Information (at least six points).

[B] Explain different ways to represent characters in a computer. [05]

[C] Explain the process of acquisition of audio type of data. [02]

OR

Q.2 [A] What do you mean by image scanner? List out types of image scanners. Explain any one type of image scanner in details. [05]

[B] Explain Capacitor Storage Cell in details. [05]

[C] Explain Backup storage with characteristics. [02]

Q.3 Answer following questions in detail (Any Three). [12]

1. Explain memory and processing system of computer.
2. Explain OCR, OMR, MICR and Bar Code Readers input units.

[P.T.O]

3. Explain the characteristics of storage cell.
4. Convert following decimal numbers in binary numbers :
 (a) 456 (b) 688
5. Explain acquisition of black and white and colour pictures.

SECTION-2

Q.4 [A] Match column A with the correct answer on column B. [05]

Column A

1. Logical Operations
2. CPU
3. Image Data Processing
4. Online Banking System
5. Copy Right

Column B

- a. Feature Extraction
- b. Literary Work
- c. Bit Operators
- d. Interpret and Execute instructions
- e. Pay Utility Bills

[B] Give an appropriate answer from following multi choice questions. [06]

1. Space Control System is an example of _____ application.
 (a) Mobile (b) Real time (c) Online (d) Desktop
2. Applications which are installed and accessible from the specific computer or laptop is _____ applications.
 (a) Mobile (b) Real time (c) Online (d) Desktop
3. _____ is not an Input/Output type of operation.
 (a) Read (b) Display (c) Load (d) Print
4. Industry 4.0 includes _____ type of system.
 (a) Cyber Physical (b) IOTs (c) Cloud Computing (d) All of these
5. If an application fails to give response in time frame and provides critical failure is considered as _____ realtime system.
 (a) General (b) Hard (c) Soft (d) None of these
6. _____ is one kind of numerical data based software.
 (a) MS Word (b) MS Excel (c) MS Project (d) MS Power Point.

Q.5 [A] Explain any six features of Spreadsheet like softwares. [06]

[B] What do you mean by online applications? Explain any five features of online shopping system. [06]

OR

Q.5 [A] Explain Electronic Data Processing(EDP) in details. [06]

[B] Explain GPS and Infrared Communications. [06]

Q.6 Answer in detail (Any Three). [12]

1. Explain Big Data Analytics and Internet of Things.
2. Explain any four societal benefits of using WWW.
3. Explain any four instruction set of CPU.
4. Explain Bluetooth technology with advantages.
5. List out any eight career opportunities of Information Technology.